11 Positron emission tomography (PET scanning) obtains diagnostic information from a person. The information is used to form an image.

(a) PET scanning uses a tracer.

Explain what is meant by a tracer.

.....

..... [1]

(b) PET scanning involves annihilation.

(i) Explain what is meant by annihilation.

.....

..... [1]

(ii) State the names of the particles involved in the annihilation process.

..... [1]

(c) **(i)** Calculate the total energy released in one annihilation event in **(b)**.

energy = ... J [1]

(ii) Calculate the wavelength of each gamma photon released.

wavelength = ... m [2]

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(d) Explain how the gamma photons are used to produce an image.

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.....

.....

..... [3]

[Total: 9]